THE PICARD FUNCTOR IS QUASI-SEPARATED AND LOCALLY OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1691-1695 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 1697-1711, the complete author proof (13 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 23 of queue_moduli.json (pb=671, stmt 1691-1695, proof 1697-1711).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Editorial note. Transfer-type lemma: the fppf-cover transfer argument is displayed in-span.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DMF: sources/stacks-moduli/pinned-original-source.tex, source lines 1676–1678. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DME: sources/stacks-moduli/pinned-original-source.tex, source lines 1639–1642. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DMB: sources/stacks-moduli/pinned-original-source.tex, source lines 1526–1529. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06Q9: sources/stacks-moduli/pinned-original-source.tex, source lines 1711–1711. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:moduli.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 06QI: sources/stacks-moduli/pinned-original-source.tex, source lines 1711–1711. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The morphism* $\mathrm{Pic}_{X/B} \to B$ *is quasi-separated and locally of finite presentation.*

Proof. To check $\mathrm{Pic}_{X/B} \to B$ is quasi-separated we have to show that its diagonal is quasi-compact. This is immediate from Lemma 0DMF. Since the morphism $\mathcal{P}ic_{X/B} \to \mathrm{Pic}_{X/B}$ is surjective, flat, and locally of finite presentation (by Lemma 0DME and Morphisms of Stacks, Lemma 06QI) it suffices to prove that $\mathcal{P}ic_{X/B} \to B$ is locally of finite presentation, see Morphisms of Stacks, Lemma 06Q9. This follows from Lemma 0DMB. □